

Introduction to Microeconomics

Lecture 0—Logistics for the course & Overview of Economics and Research in Economics

Prof. Alejandro Melo-Ponce



NAZARBAYEV
UNIVERSITY

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Who I am, where I am, and how can you find me?

- Office 8.510, however, sometimes I prefer to meet students in one of the coffee places around the University.
- Office hours by apointment, book slots by [youcanbookme](#): the link is in the syllabus. Every week I open some slots which you can book.

- Disregard the Registrar's Tutorial sections!
- I coordinate new times with the help of my TAs.
- Announcements about this will be made by the end of this week.

There is no official textbook for the course. However, the following references might be useful:

- Shapiro, David. *Principles of Macroeconomics* 3e. OpenStax College, 2022.
- Börgers, Tilman. *Lecture Notes on Microeconomic Theory*.
- Perloff, Jeffrey M. *Microeconomics*. Pearson, 2018, 8th ed.
- Frank, Robert; Bernanke, Ben; Antonovics, Kate; Heffetz, Ori. *Principles of Microeconomics*. McGraw Hill 8th ed.

Activity	Date	Weighting
Midterm Exam 1	19.09.24	30%
Midterm Exam 2	31.10.24	30%
Final	TBA	40%

A	90–100	A–	85–89.99
B+	80–84.99	B	75–79.99
B–	70–74.99	C+	65–69.99
C	60–64.99	C–	55–59.99
D+	50–54.99	D	45–49.99
F	0–44.99		



What topics does Economics study?

- A large variety of topics!
- Almost all domains of human activity can be studied with Economics!
- Economics research is organized into fields.
- The goal today is to imprint on you the role that our introductory course in microeconomic theory relates to the universe of contemporary economics research.

- Let's focus on the article by Antràs and coauthors.
- Focuses on firm's choices of which inputs for their production to import from abroad.
- Why do you think this is relevant?
- What do they find?

- The conventional structure for a research article is to have an introduction, and at the end of this introduction, there is typically a paragraph that explains how the paper is structured.
- Quoting from the article:

*The rest of the paper is structured as follows. We present the assumptions of our **model** in Section I, and solve for the equilibrium in Section II. In Section III we introduce the **data** and provide descriptive evidence supporting the assumptions underlying our theoretical framework. We estimate the model structurally in Section IV, and in Section V, we perform our counterfactual analysis and compare the predictions of the model to reduced-form evidence. Section VI concludes.*

- The data comes from US Customs and Borders Protection.
- It also includes data from other US government agencies.
- Some of it is confidential.

Consider a world consisting of J countries in which individuals value the consumption of differentiated varieties of manufactured goods according to a standard symmetric CES aggregator

$$U_{M_i} = \left(\int_{\omega \in \Omega_i} q_i(\omega_i)^{(\sigma-1)/\sigma} d\omega \right)^{\sigma/(\sigma-1)} \quad (1)$$

- A **model** is something mathematical.
- A mathematical description of how that aspect of the economy works:
 - How inputs are transformed into outputs, how decisions about inputs and outputs are made, how much of the output firms will be able to sell, how this depends on the prices firms choose, how firms will set their prices, etc.
 - Specifically, for equation (1), it describes how much enjoyment people derive in the model from consumption of manufactured goods.
 - The LHS is a symbol for “enjoyment” (**utility** in the jargon of Economics).
 - The RHS is a formula that involves quantities of the various goods a person buys.
 - In the model, the utility will determine the demand for manufactured goods.
 - The demand, in turn, contributes to the decision that firms make about which inputs to buy from abroad.
- But, how can a mathematical description of manufactured goods, imports, prices, come **before we see the data?**

- Let's take another look:

$$U_{M_i} = \left(\int_{\omega \in \Omega_i} q_i(\omega_i)^{(\sigma-1)/\sigma} d\omega \right)^{\sigma/(\sigma-1)}$$

- What is the value of σ ?
- That is what the authors want to figure from the data!
- After invoking their data, they estimate the value to be $\sigma = 3.85$.

- Before even seeing the data, the authors make assumptions about the consumers' utility, as expressed by (1).
- Then, they look at the data assuming that this formula is correct, leaving the only remaining question to be what is the value of σ .
- There are many other assumptions: how firms transform inputs into outputs, the market structure, etc.
- The model is just a **lens or filter** through which the authors look at their data, and this lens, together with what is out there, determine what the authors focus on.

Why to use a model then?

- Don't lenses distort what you see?
- Models allow economists to focus on a particular aspect of what they see.
- In this example, it permits the authors to focus on what they see in their data, and prevents them from getting lost in the sea of data.
- It allows them to **focus and what they want to investigate**.
- Another benefit: a model allows researchers to ask questions about economic policy: if the model and derivations are correct, we can test new government policies that don't yet exist within the model to simulate their effects.

Does the model fit perfectly the data?

- Of course not, that would be miraculous.
- There are random effects not captured by the model.
- To cope with these, the authors use methods from statistics.
- The specific methods that we have developed to analyze economic data is called **econometrics**.

- ①
- ② Find an interesting question.
- ③ Develop a mathematical model that is suitable.
- ④ Use data to figure out the value of the parameters.
- ⑤ Use the understanding of the world developed in this exercise to make predictions.

Do all economic research look like this?



What is the role of Microeconomic Theory

- Two skills that you will develop as an Economist:
 - The ability to write suitable mathematical models.
 - The econometric skills needed when taking models to real world data.
- Two fields concerned with developing the tools and ideas used in these two skills:
 - Theory
 - Econometric Theory
- Research in these fields often looks like pure mathematics rather than what a layperson would consider “economics”.
- This class is an introduction to the first set of skills:
 - You will be introduced to simplified versions of the main tools that appear in Economic Models.
 - Think of Economic Models as a **fable**: in this class you will meet the simple version of the characters that populate economic fables.
 - The models might look too simple or naive, you might feel that they are too disconnected from the real world. But even in this setting you will soon encounter things that look similar to (1) and hopefully you will soon understand how such things are used in Economics.

- Macroeconomics is the study of fluctuations in the total value of production in an economy, and with the growth and shrinkage of that production.
- Is there a difference between macroeconomic and microeconomic theory?
- The distinction is old fashioned (1960's).
- Modern macroeconomic theory¹ uses tools from microeconomic theory.

¹(That is, good macroeconomic theory! Caveat: this is a personal opinion, don't tell your local macroeconomists unless you have more information about their preferences!)